Non-uniqueness and symmetries for the Nirenberg problem using computer assistance

Daniel Platt (Imperial College London)
DANGER, BIRS, 9 Apr 2026

The Nirenberg problem asks: which functions are the curvature of a metric on a sphere that is conformal to the round metric? This reduces to solving a non-linear partial differential equation (PDE) for a function on the sphere. In the recent arXiv:2602.12368, machine learning was used to conjecture that certain curvatures admit a solution to the PDE, based on numerics. In the talk, I will explain how to upgrade one of these conjectural examples to a proven example using a numerically verified proof. The general strategy can be applied to many partial differential equations and I will highlight the parts that easily carry over to other cases and the parts that require hard work in each case.

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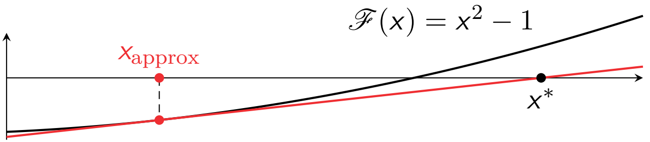
Example 1: Baby example

- Zero of $\mathscr{F}(x) = x^2 - 1$. Start with x_{approx}. Newton iteration: does it converge?



- Computer: $x_{approx} = 0.999$. Write $x = x_{approx} + v$. Taylor:

$$\mathscr{F}(x) = \mathscr{F}(x_{approx}) + \mathscr{L}(v) + \mathscr{N}(v), \quad \mathscr{L} \text{ linear}, \quad \mathscr{N} \text{ h.o.t.}$$

- Error: $|\mathscr{E}| := |\mathscr{F}(x_{approx})| \leq 0.002$

Theorem (Newton-Kantorovich)

Let $r > 0$ such that for $u_1, u_2 \in B_r := \{u \in \mathbb{R} : |u| \leq r\}$ we have

$$|\mathscr{E}| \leq c_1, ||\mathscr{L}^{-1}|| \leq c_2, |\mathscr{N} u_1 - \mathscr{N} u_2| \leq c_3(|u_1| + |u_2|)|u_1 - u_2|$$

and $c_2 c_1 + c_2 c_3 r^2 \leq r$, $2 c_2 c_3 r < 1$, then ex. $v \in B_r$ such that $\mathscr{F}(u_0 + v) = 0$.

- For $x_{\mathrm{approx}} = 0.999$: $c_1 = 0.002$, $c_2 = 0.501$, $c_3 = 1$ ✅ for $r = 0.002$

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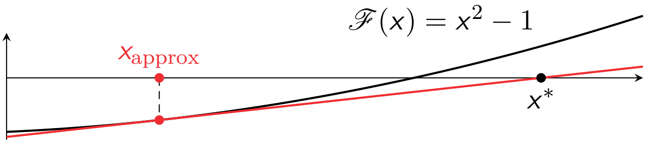

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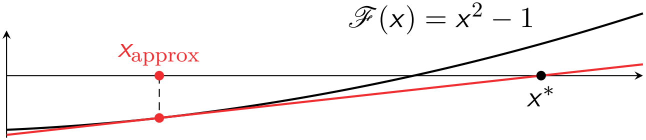

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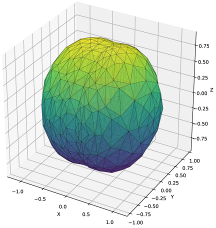

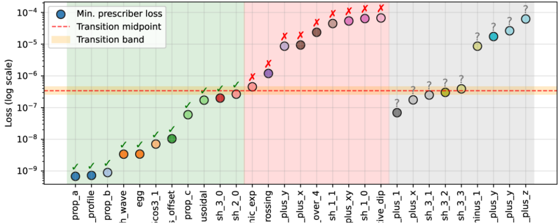

Images from "A Machine Learning Approach to the Nirenberg Problem"

- Equivalent: does $1 - \Delta u - Ke^{2u} = 0$ have solution $u : S^2 \to \mathbb{R}$?
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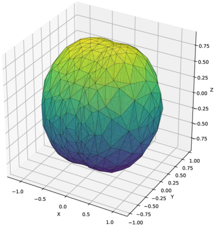

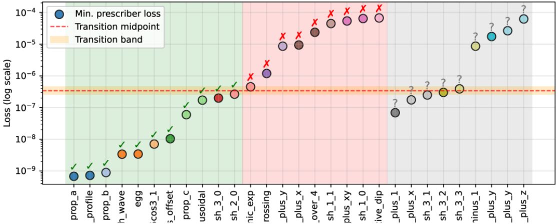

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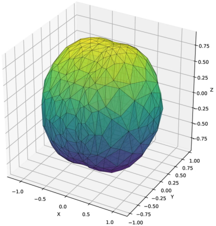

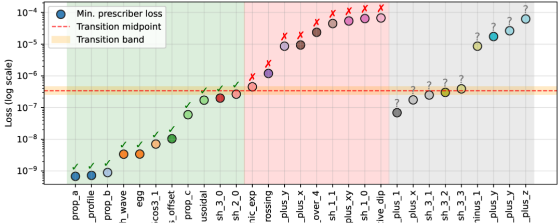

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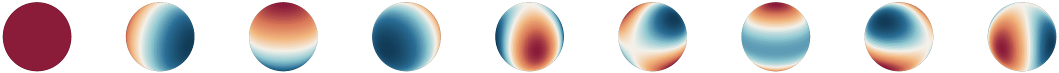
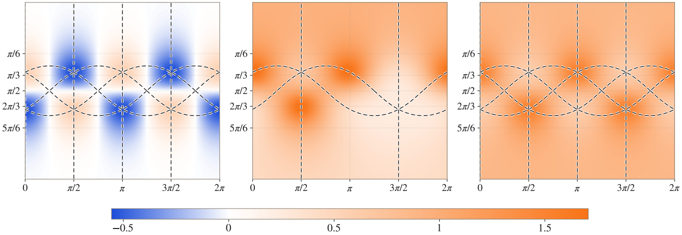


- $Y_{3,2}$ has symmetry T_d (tetrahedral group) with $|T_d| = 24$

Theorem

For $K = Y_{3,2}$ there are at least five solutions to the equation $1 - \Delta u - Ke^{2u} = 0$. One has symmetry group T_d, four have symmetry group S_3.



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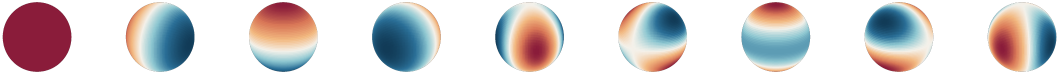

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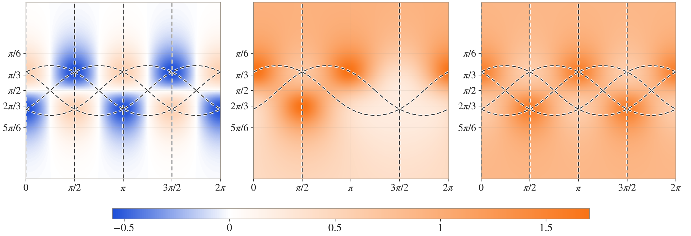

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- Approximate solution $u_0 \in H^2$, i.e. $\mathscr{E} := \mathscr{F}(u_0)$ small
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- Compute some $u_0 = \sum_{0 \leq l, |m| \leq 50} a_{lm} Y_{l,m}$ (e.g. neural network)
- Bound $||u_0||_\infty \leq \sum |a_{lm}| \cdot ||Y_{l,m}||_\infty < 1.58$ not sharp (🤥 really Lipschitz)
- For $p = 50$ let $\exp_p$ degree p Taylor poly of exp, then: $\left|\left|\exp(2u_0) - \exp_p(2u_0)\right|\right|_\infty \lesssim \frac{||u_0||_\infty}{(p+1)!}$ small
- $||\mathscr{E}||_{L^2} \leq \left|\left|1 - \Delta u_0 - K\exp_p(2u_0)\right|\right|_{L^2} + \left|\left|K(\exp(2u_0) - \exp_p(2u_0))\right|\right|_{L^2} < 2.2 \cdot 10^{-8}$

  Where first summand computed in spherical harmonics coefficient space

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and $c_2 c_1 + c_2 c_3 r^2 \leq r$, $2 c_2 c_3 r < 1$, then ex. $v \in B_r$ such that $\mathscr{F}(u_0 + v) = 0$.

- Compute some $u_0 = \sum_{0 \leq l, |m| \leq 50} a_{lm} Y_{l,m}$ (e.g. neural network)
- Bound $||u_0||_\infty \leq \sum |a_{lm}| \cdot ||Y_{l,m}||_\infty < 1.58$ not sharp (🤥 really Lipschitz)
- For $p = 50$ let $\exp_p$ degree p Taylor poly of exp, then: $\left|\left|\exp(2u_0) - \exp_p(2u_0)\right|\right|_\infty \lesssim \frac{||u_0||_\infty}{(p+1)!}$ small
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Ex 2: Proof step 1: bound the residual $\mathscr{E}$

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$$\mathscr{L} = \begin{pmatrix} A & C \\ C^* & D \end{pmatrix}$$

- A : $||Au||_{L^2} \geq 2.04\,||u||_{L^2}$ fin.-dim. computation (🤥 tricky linear algebra)
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Ex 2: Proof step 3: Higher order terms and conclusion

- $\mathcal{N}(u) := u^2$ (🤥 omitted some terms)

Theorem (Newton-Kantorovich)

Let $r > 0$ such that for $u_1, u_2 \in B_r := \{u \in H^2 : ||u|| \le r\}$ we have

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and $c_2 c_1 + c_2 c_3 r^2 \le r$, $2c_2 c_3 r < 1$, then ex. $v \in B_r$ such that $\mathcal{F}(u_0 + v) = 0$.

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- Check $c_2 c_1 + c_2 c_3 r^2 \le r$, $2c_2 c_3 r < 1$ satisfied for $r = 3.1 \cdot 10^{-4}$ $\rightsquigarrow$ $\hat{u}$ exists

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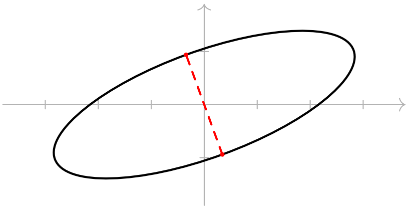
Example 3: bottlenecks of submanifolds [Tin23]

- $\gamma : I \to \mathbb{R}^2$ curve
- $p = \gamma(t_1), q = \gamma(t_2)$ bottleneck if critical point for distance function



- E.g. $\gamma(t) = \begin{pmatrix} 3\cos t \cos 20° - \sin t \sin 20° \\ 3 \cos t \sin 20° + \sin t \cos 20° \end{pmatrix}$
- $D(s,t) = \frac{1}{2}|\gamma(t) - \gamma(s)|^2_{\mathrm{Eucl}}, \qquad \mathscr{F}(s,t) = \nabla D(s,t)$
- $x_{\mathrm{approx}} = (s_0, t_0) = (1.571,\ 4.712),\ \mathscr{E} = \mathscr{F}(x_{\mathrm{approx}}) \rightsquigarrow |\mathscr{E}|_\infty \leq 2.1 \cdot 10^{-3}$
- $\mathscr{L} = d\mathscr{F}_{x_{\mathrm{approx}}} = \mathrm{Hess}\, D(x_{\mathrm{approx}}),\ |\mathscr{L}^{-1}|_\infty \leq 0.501$
- $|\mathscr{N}(v_1) - \mathscr{N}(v_2)|_\infty \leq 56\,(|v_1|_\infty + |v_2|_\infty)\,|v_1 - v_2|_\infty$
- Thus $c_1 = 2.1 \cdot 10^{-3},\ c_2 = 0.501,\ c_3 = 56.$ For $r = 1.1 \cdot 10^{-3}$ have $c_2 c_1 + c_2 c_3 r^2 < r, \qquad 2 c_2 c_3 r < 1 \rightsquigarrow$ bottleneck exists near x_{approx}

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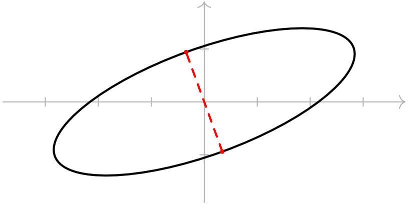

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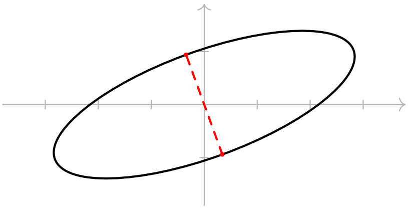

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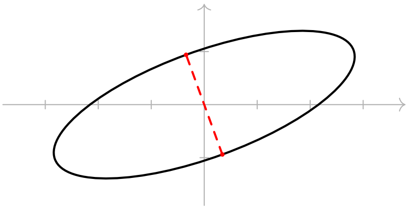
Example 3: bottlenecks of submanifolds [Tin23]



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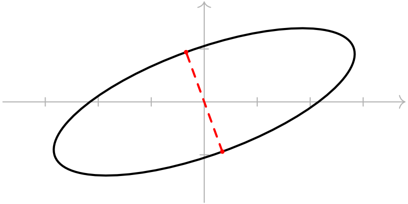

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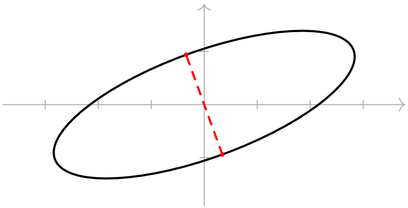

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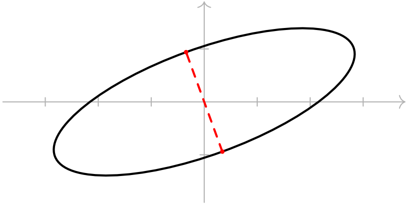

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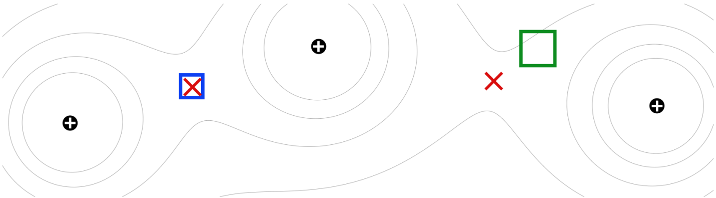

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- Step 1: certify solutions using Newton-Kantorowich as before (blue box)
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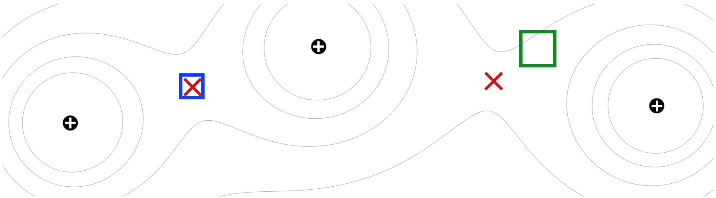

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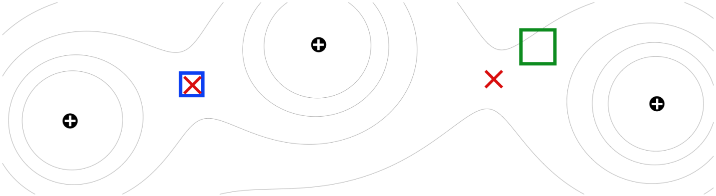

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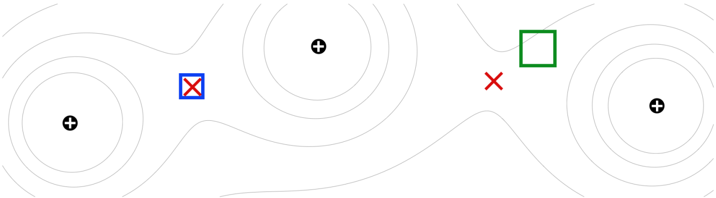

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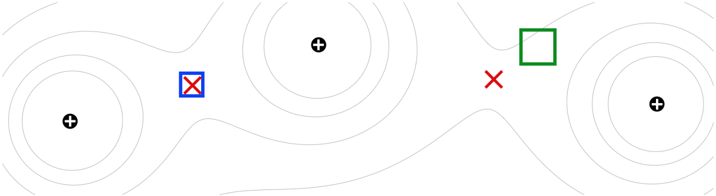

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Thank you for the attention!

References I

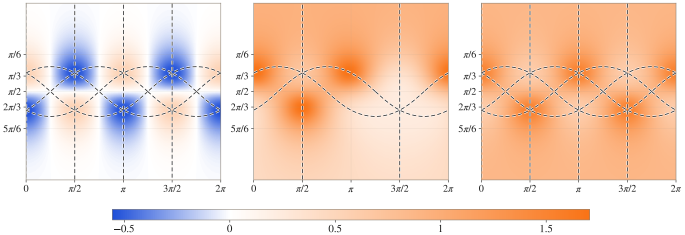
Herbert Edelsbrunner, Christopher Fillmore, and Gonçalo Oliveira.
Counting equilibria of the electrostatic potential.
arXiv preprint arXiv:2501.05315, 2025.

Raphaël Tinarrage.
Recovering the homology of immersed manifolds.
Discrete & Computational Geometry, 69(3):659–744, 2023.

Symmetries

Theorem

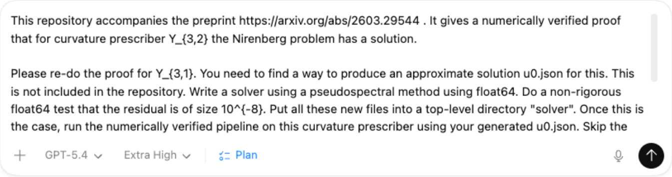
For $K = Y_{3,2}$ there are at least five solutions to the equation $1 - \Delta u - Ke^{2u} = 0$. One has symmetry group T_d, four have symmetry group S_3.



- Start with u_0 exactly T_d/S_3-invariant
- Define L^2 and H^2 to be only invariant functions;
  $\mathscr{L}, \mathscr{N}$ also invariant $\Rightarrow \hat{u}$ invariant
- For S_3 case rule out extra symmetries: $g \in T_d \setminus S_3$, find x s.t.
  $|u_0(x) - g^*(u_0)(x)| > 0.1$, then
  $|\hat{u}(x) - g^*(\hat{u})(x)| \geq |u_0(x) - g^*(u_0)(x)| - 2\,||\hat{u} - u_0||_\infty \geq 0.1 - 10^{-2} \neq 0$
  $\Rightarrow g^*(\hat{u}) \neq \hat{u}$ (🤥 extra trick: symmetries at most T_d)

AI coding tool usage

- AI agent can write the whole thing in one go (e.g. OpenAI Codex)



- Problem: completely unreviewable, non-rigorous arithmetic in several places
- Useful for prototype/viability (identify too slow computations)
- Give code structure (methods/classes): fewer non-rigour errors and easy to find
- Produce unit tests to find potential issues (needn't be trusted)